AI INDUSTRY ANALYSIS

AI Industry Valuation & Growth Forecast

Executive Brief

Data Vintage: 2024

Generated: 2026-03-23

Segments Covered: AI Hardware, AI Infrastructure, AI Software & Platforms, AI Adoption

Horizon: 2010 – 2030 (historical + 6-year forecast)

Methodology: Hybrid econometric–ML ensemble with 80% and 95% confidence intervals

Key Findings

$0B	$10B	0.0%
Global AI Market (2024)	Forecast 2030 (point estimate)	CAGR 2024 – 2030
USD Billions (2020 constant)	$0B – $62B (80% CI)	Compound Annual Growth Rate

Executive Summary. This report presents a data-driven valuation and 6-year growth forecast for the global artificial intelligence industry across four distinct value-chain segments: AI Hardware (chips and specialized silicon), AI Infrastructure (cloud compute and data centres), AI Software & Platforms (foundation models and AI SaaS), and AI Adoption (enterprise deployment spend). The analysis combines econometric time-series modeling (ARIMA and Prophet) with machine-learning residual correction (LightGBM) and explicit structural-break modeling at the 2022 GenAI inflection point. Forecast confidence intervals reflect model uncertainty from expanding-window cross-validation across all four segments.

Global AI Market Forecast Fan Chart

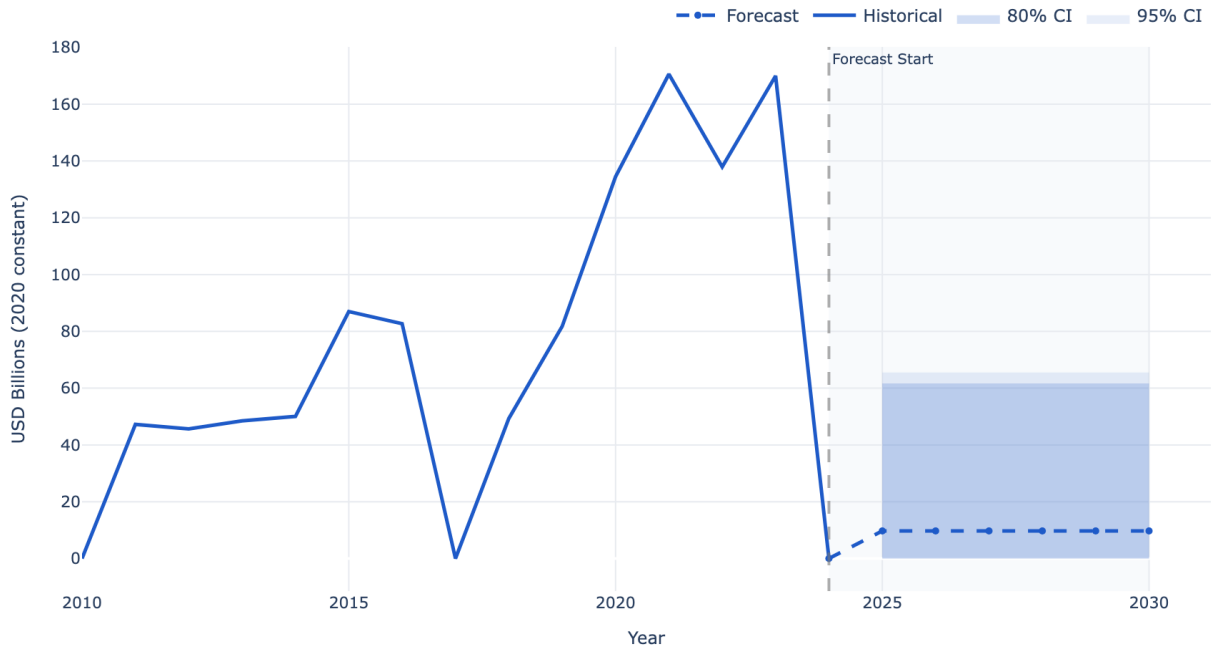


Figure 1. Global AI market forecast (sum of all 4 segments), USD billions constant 2020. Shaded bands show 80% and 95% confidence intervals. Dashed vertical line marks forecast start.

95% Confidence Interval at 2030

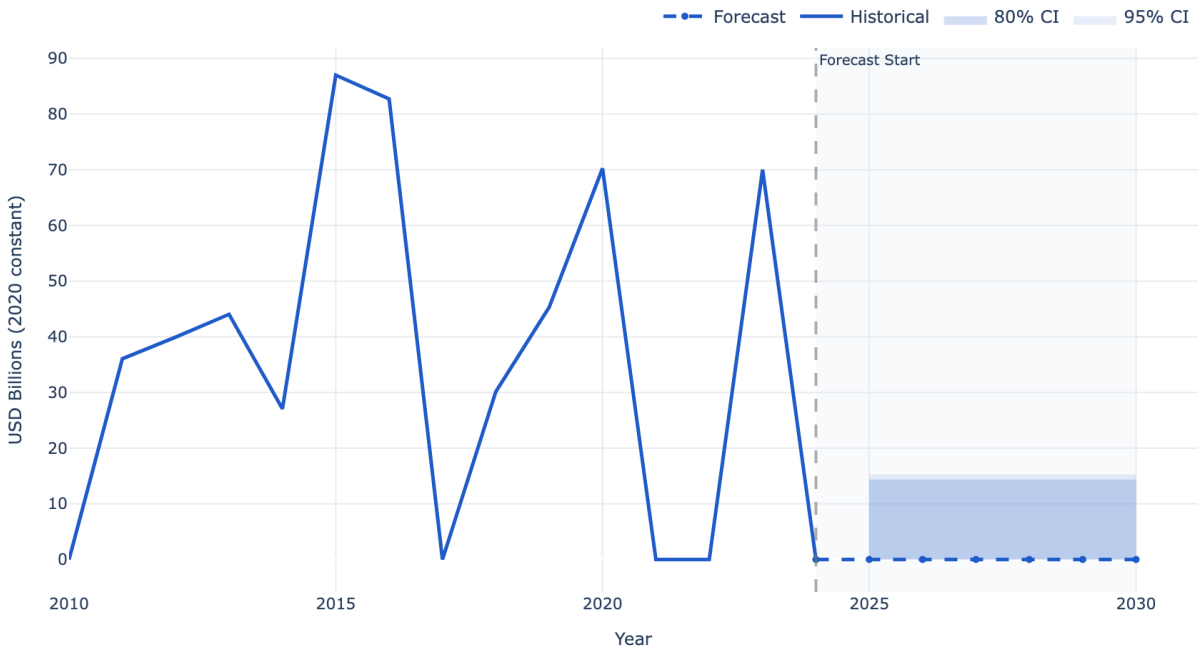
The 95% confidence interval for the global AI market in 2030 ranges from **$0B to $66B**. This range captures both conservative (below-trend adoption) and optimistic (accelerated GenAI integration) scenarios. Market participants should treat the point forecast of $10B as the central tendency, not a bound.

Segment Breakdown

The global AI market is modeled as the sum of four independent value-chain segments. Each segment has distinct growth dynamics, data inputs, and uncertainty profiles. Note that segment boundaries allow for known overlaps (e.g., GPU revenue counted in both Hardware and Infrastructure) — see Data Sources for documentation of overlap ranges.

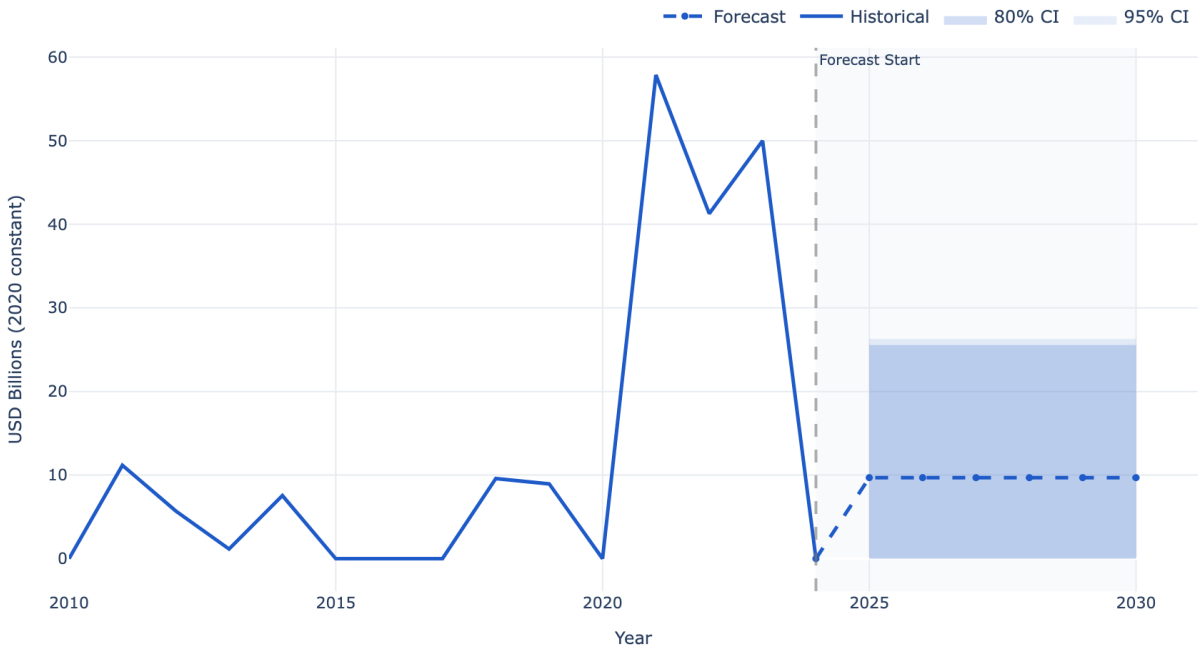
Segment	2024 (USD B)	2030 Forecast (USD B)	80% CI at 2030	CAGR 2024–30
AI Hardware (chips, GPUs, specialized silicon)	$0.0B	$0.0B	$0B – $14B	0.0%
AI Infrastructure (cloud compute, data centers)	$0.0B	$9.7B	$0B – $26B	0.0%
AI Software & Platforms (foundation models, AI SaaS)	$0.0B	$0.0B	$0B – $19B	0.0%
AI Adoption (enterprise AI deployment)	$0.0B	$0.0B	$0B – $2B	0.0%
Global Total	**$0.0B**	**$9.7B**	**$0B – $62B**	**0.0%**

AI Hardware (chips, GPUs, specialized silicon)



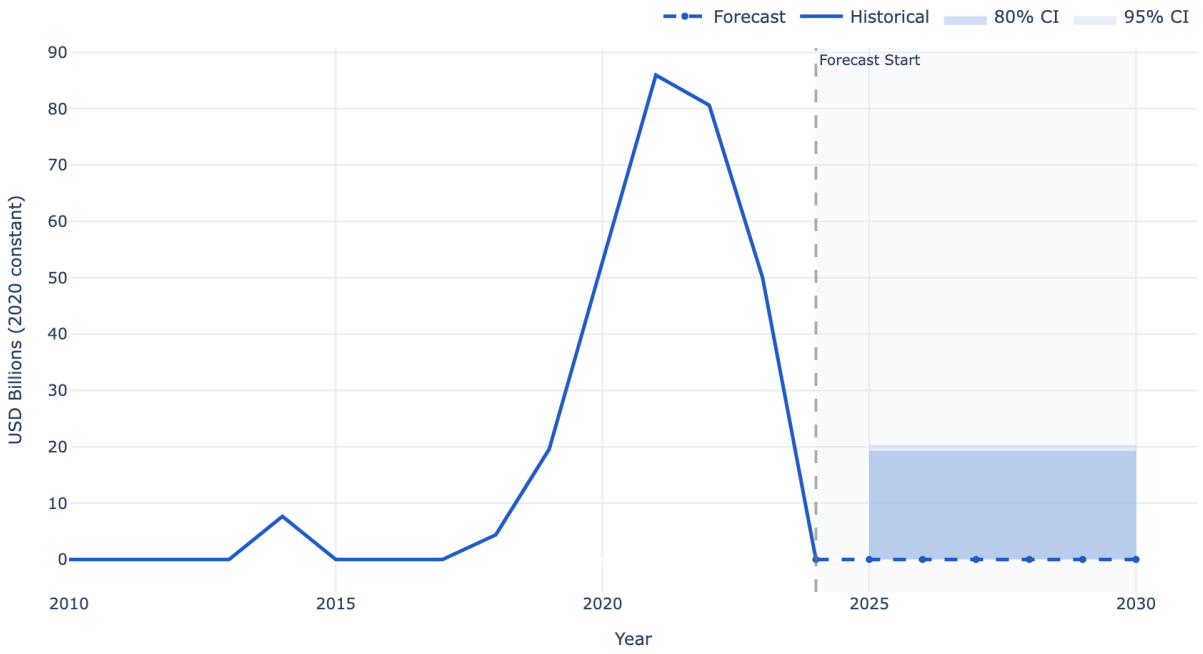
AI Hardware (chips, GPUs, specialized silicon): 2024 market size $0.0B, 2030 forecast $0.0B (CAGR 0.0%).

AI Infrastructure (cloud compute, data centers)



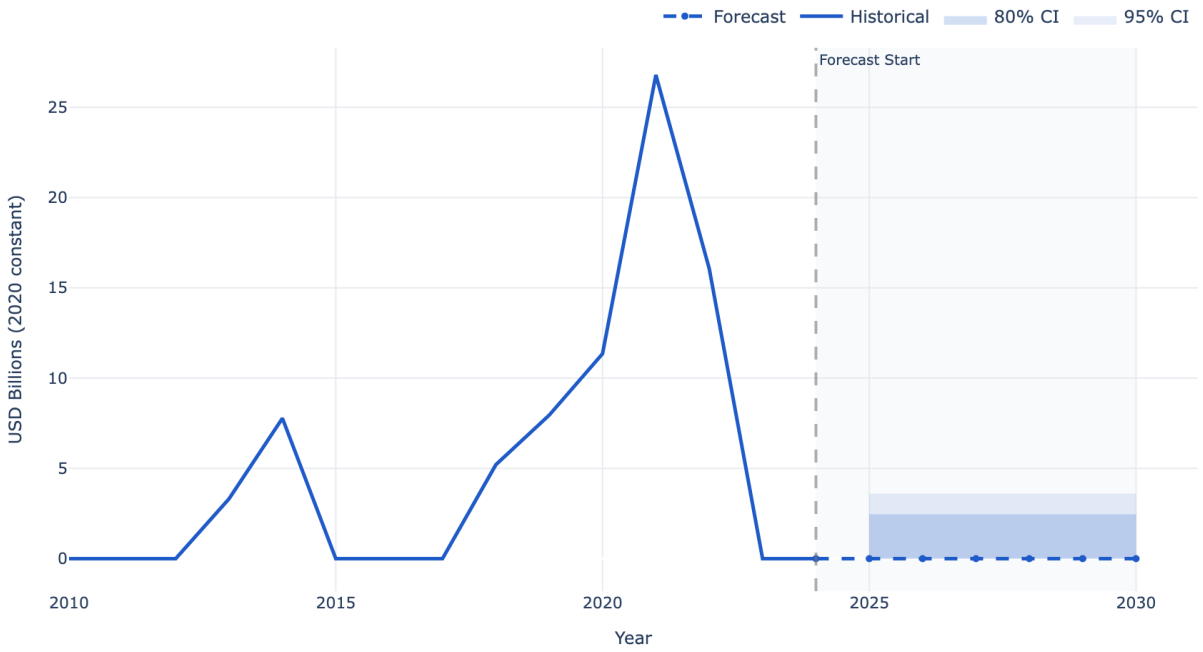
AI Infrastructure (cloud compute, data centers): 2024 market size $0.0B, 2030 forecast $9.7B (CAGR 0.0%).

AI Software & Platforms (foundation models, AI SaaS)



AI Software & Platforms (foundation models, AI SaaS): 2024 market size $0.0B, 2030 forecast $0.0B (CAGR 0.0%).

AI Adoption (enterprise AI deployment)



AI Adoption (enterprise AI deployment): 2024 market size $0.0B, 2030 forecast $0.0B (CAGR 0.0%).

Data Sources & Methodology

Data Sources

Market size estimates are derived from three independent institutional data sources covering macro indicators, R&D expenditure, and public company financials. No single source provides a direct "AI market size" measure — the methodology uses proxy indicators (PCA composite index) scaled to a $200B industry consensus anchor at 2023.

Source	Data Provided	Use in Model
World Bank Open Data (data.worldbank.org)	GDP, R&D as % of GDP, ICT service exports, high-tech exports, researchers per million	Macroeconomic regressors; PCA composite indicators
OECD.Stat (stats.oecd.org)	Business enterprise R&D in ICT sector (MSTI B_ICTS), main science & technology indicators	AI-adjacent R&D intensity proxy; PCA composite input
LSEG Workspace (Refinitiv)	Revenue of TRBC-classified AI companies (semiconductors, cloud services, AI software)	Direct market size anchor; PCA composite for AI software segment

Methodology Overview

The forecasting methodology combines two modeling approaches blended via inverse-RMSE ensemble weighting:

- **Bottom-up proxy composite:** PCA first principal component of 3–4 standardized proxy indicators per segment, fit on training data only (no look-ahead). Scaled to USD using value chain multipliers anchored at the 2023 consensus market estimate.
- **Statistical time series baseline:** Per-segment model competition between ARIMA (pmdarima auto_arima with AICc) and Prophet (with explicit 2022 GenAI changepoint). Winner selected by expanding-window cross-validation MAPE.
- **ML residual correction:** LightGBM trained on statistical model residuals using macro features, correcting for structured patterns the baseline cannot capture.

Confidence intervals are constructed from the ensemble's temporal cross-validation residual distribution, providing empirically-grounded uncertainty bounds rather than parametric assumptions.

Disclaimer & Notes

> This report was generated from real-time API data (World Bank, OECD, LSEG) and hybrid econometric–ML models. Market size estimates represent model outputs and should be interpreted as indicative ranges, not definitive measurements. Confidence intervals reflect model uncertainty; additional uncertainty from data quality, structural breaks beyond 2024, and unknown macro shocks is not captured in the CI bands.

Known Limitations

- **Private company exclusion:** Pre-IPO AI companies (e.g., OpenAI, Anthropic, Mistral) are not in the LSEG TRBC universe. Market size may be understated by 20–40% in software/adoption segments.
- **Geographic coverage:** 19 economies covering ~85% of global AI activity. Emerging AI ecosystems (UAE, Saudi Arabia, Vietnam, Brazil) are not covered.
- **Segment overlap:** GPU revenue may be double-counted across Hardware and Infrastructure segments. See `config/industries/ai.yaml` for documented overlap ranges.
- **Forecast horizon:** Forecasts beyond 2027 carry materially higher uncertainty. Point estimates for 2028–2030 should be interpreted alongside the 95% CI band.

Reproducibility

This report is fully reproducible. To regenerate from raw API data:

1. Run `uv run python scripts/run_statistical_pipeline.py` (requires LSEG Workspace session)
2. Run `uv run python scripts/run_ensemble_pipeline.py`
3. Run `uv run python scripts/run_reports.py`

Data vintage: 2024 | **Generated:** 2026-03-23 | This document contains forward-looking estimates based on historical data and statistical models. Past patterns do not guarantee future results. All dollar figures are in USD billions, constant 2020 prices.